

Tutorial 8: Optimal Cryptocurrency Design and Welfare

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Source. The problems below are based on Chiu and Koepl (2017), *The Economics of Cryptocurrencies – Bitcoin and Beyond* (cited throughout as CK2017), Sections 4.6 and 5. Tutorial 7 covered the mining game and the double-spending constraint; here we use those building blocks to study optimal design, calibration, and welfare.

Learning objectives. After completing this tutorial you should be able to

- state and explain the planner’s Ramsey problem for cryptocurrency design;
- explain why optimal design uses seigniorage rather than transaction fees;
- compute consumption-equivalent welfare costs and interpret them in context;
- contrast cryptocurrencies with cash and with centralised payment systems (Fedwire, debit cards);
- articulate the link between scalability and welfare, and the implications for policy debates around CBDCs.

1. **The planner’s problem.** The cryptocurrency designer chooses (μ, τ) to maximise social welfare

$$\mathcal{W} = B\sigma \int [\delta^{N(\varepsilon)} \varepsilon u(x(\varepsilon)) - x(\varepsilon)] dF_\varepsilon(\varepsilon) - C(\bar{N} + 1),$$

subject to: (i) $(d(\varepsilon), x(\varepsilon), N(\varepsilon))$ is the buyer’s optimal contract given (μ, τ) ; (ii) DS-proof constraint $d(\varepsilon) \leq R(N(\varepsilon) + 1)N(\varepsilon)$; and (iii) the planner’s budget $\frac{\beta}{\mu}R(\bar{N} + 1) = B(\mu - 1 + \sigma\tau) \int \frac{x(\varepsilon)}{1-\tau} dF_\varepsilon(\varepsilon)$.

- (a) Why does the planner have to take the agents’ equilibrium choices as a constraint? Compare this with a Ramsey optimal-tax problem.
 - (b) Explain in words why the budget constraint takes this particular form, and where the factor $\frac{\beta}{\mu}$ on the left-hand side comes from.
 - (c) Why is the cost of mining represented as $C(\bar{N} + 1)$ and how does Lemma 1 of CK2017 allow us to substitute $C(\bar{N} + 1) = \frac{\beta}{\mu}R(\bar{N} + 1)$?
2. **Optimal mix of seigniorage and fees.** CK2017’s Proposition 8 establishes that the optimal design sets $\tau = 0$ and $\mu > 1$.

- (a) Provide an intuitive argument for why financing mining rewards through seigniorage *dominates* financing them through transaction fees. Your argument should mention the incidence of each instrument across buyers with different valuations.
 - (b) Why does the planner not want to set $\mu = 1$ (the Friedman rule)? Identify the marginal benefit and the marginal cost of raising μ above one.
 - (c) Why might a private cryptocurrency network (e.g. Bitcoin) end up with a positive τ even though the optimal τ is zero? Discuss in terms of governance and short-run incentives of miners.
3. **Welfare cost: Bitcoin vs. cash.** The benchmark calibration of CK2017 yields a Bitcoin welfare cost of 1.41% of consumption, falling to 0.08% under optimal design. Roughly, an equivalent low-inflation cash system generates a welfare cost of only 0.003%.
- (a) Define carefully the two welfare measures used in Section 5.3: the *consumption-equivalent welfare loss* and the *equivalent inflation tax*.
 - (b) Bitcoin’s mining cost is calibrated at $\approx$ \$360 million per year. Explain in one paragraph why mining cost dominates the welfare comparison.
 - (c) Compute (qualitatively) how much of the 1.41% $\rightarrow$ 0.08% welfare improvement under optimal design comes from (i) lowering μ , (ii) setting $\tau = 0$, and (iii) reducing confirmation lags. Which is quantitatively the most important?
 - (d) Suppose the social planner were forced to keep τ at Bitcoin’s calibrated level (0.0088%) but could freely choose μ . Sketch how the optimal μ and welfare cost would change.
4. **Scale and the security–volume nexus.** A key feature of cryptocurrencies in CK2017 is that mining rewards scale with aggregate transaction volume, while double-spending incentives scale with individual transaction size.
- (a) Explain why this asymmetry implies that cryptocurrencies become *more efficient* as the user base B grows. Connect your argument to the DS constraint $d < R(N + 1)N$ and the formula $R = [Z(\mu - 1) + D\tau]/(\bar{N} + 1)$.
 - (b) Use the asymmetry to predict whether a cryptocurrency is better suited to support a retail system (e.g. debit cards) or a large-value system (e.g. Fedwire). Reconcile your prediction with CK2017’s Table 5.4: welfare loss 0.00052% (retail) vs. 0.0060% (large value).
 - (c) Using the table, compare CK2017’s implied break-even transaction fee for an optimally-designed cryptocurrency with the observed debit-card interchange fee ($\approx$ 23 cents per transfer) and the Fedwire fee ($\approx$ 82 cents). Which payment system can cryptocurrencies plausibly compete with, and why?
5. **Scaling and the finality trade-off.**
- (a) Recall from Tutorial 7 that settlement cannot be both immediate and final. How does this trade-off interact with the choice of μ in the planner’s problem? (Hint: higher μ relaxes the DS constraint and shortens optimal N , but raises inflation.)
 - (b) CK2017 abstract from technological scalability constraints (block size, network propagation). Explain how a binding scaling cap would alter the welfare comparison between a cryptocurrency and a centralised retail payment system.

- (c) Bitcoin processes ≈ 7 transactions/second, while Visa sustains $\approx 1,700$ /second. What design margins (block size, block interval, off-chain settlement) could relax this constraint, and what new trade-offs would they introduce?
6. **Policy: central bank digital currencies (CBDC).** Several central banks (Bank of Canada, ECB, RBA) are exploring CBDCs.
- (a) A CBDC is centrally issued, so the double-spending problem can be solved without PoW mining. Using the framework of CK2017, what does this imply about the welfare cost of a CBDC relative to (i) Bitcoin and (ii) a cash system under the Friedman rule?
- (b) Suppose a central bank wishes to use the inflation tax on a CBDC to finance a public good (say, a payments security infrastructure). Discuss whether the analogue of CK2017's Proposition 8 ($\tau = 0$, $\mu > 1$) would still hold, and under what assumptions it would fail.
- (c) Some authors (Agarwal–Kimball, Rogoff) argue that digital currency adoption could facilitate negative interest rate policy or curtail tax evasion. How would the CK2017 framework need to be extended to evaluate these arguments?

References

- [1] Chiu, J. and T. V. Koepl (2017). The Economics of Cryptocurrencies – Bitcoin and Beyond. *Bank of Canada Staff Working Paper / Queen's University*.
- [2] Lagos, R. and R. Wright (2005). A Unified Framework for Monetary Theory and Policy Analysis. *Journal of Political Economy* 113(3), 463–484.
- [3] Ron, D. and A. Shamir (2013). Quantitative Analysis of the Full Bitcoin Transaction Graph. *Financial Cryptography*.